

Go through the instructions on how to study :

1.

<https://drive.google.com/file/d/0Byt7-j-JD0d0bmxlRkZGcjN2cjA/view?resourcekey=0-VfnVIJ2CEXaqv-0JK6w4CQ>

2. file:///home/ubuntu/Desktop/MY_WORK/GATE%202027%20/SAPATH_GATE_CS_2027_MASTER_PLAN_v9_ELITE%20(1).pdf

PRACTISE MATERIAL FOR ALL SUBJECTS

Simple workflow : Lectures , PYQ (from PYQ lectures) , Revise , again and again , until you can solve all PYQ's and other problems easily without looking at the lectures or solutions easily .

PRACTISE ENOUGH , PRACTISE AGAIN AND AGAIN , NOT A LOT , BUT SAME AGAIN AND AGAIN

Be urgent , have a sense of urgency working all the time , because it might seem like you have a lot of time , but you don't

1. GATE OVERFLOW 2027 SERIES :

<https://github.com/GATEOverflow/GO-PDFs/releases/tag/gatecse-2027>

2. COVER ALL THESE PDF's without failure : <https://github.com/GATEOverflow/GO-PDFs>

LOOK FOR BOOKS ONLY WHEN YOU NEED MORE CONCEPTUAL CLARITY REGARDING SOME TOPICS . LEAVE IF YOU DON'T

3. Github that contains **PDF of all books** :

<https://github.com/gigahidjrikaaa/Engineering-Books/tree/main>

4. Book list suggestion : <https://gatecse.in/best-books-for-gatecse/>

5. <https://drive.google.com/drive/u/0/folders/1GQIPcmeRrUn2BCesfxK2re33Cp7rHz-v>

C PROGRAMMING , DATA STRUCTURES AND ALGORITHMS :

Types of Problems

• Understand Different Problems on Stack, Queue, Link List. Generally they come in a C program.

- Properties of Heap. Deletion and insertion of items in the heap.
- Practice Tree problems like no of leaf nodes, non leaf nodes, total nodes, height of the tree, no of full nodes, mirror image, etc. AVL tree and balancing them on insertion and Deletion. Binary tree, Binary Search Tree, Inorder, Preorder, Postorder traversal.
- Spanning Trees, Minimum Spanning Tree problems.
- Finding Complexity : Sometimes direct questions are asked say find complexity of Heap sort, but mostly you are given a code or question. You need to find best / average case complexity of that problem. So try to find complexity of every algorithm or program which you practice. Understand properties of complexity. Sometimes relation between them is asked.
- Searching and Sorting Problems. Difference between Different Techniques and how to apply them on different real life problems.
- Questions on approach of dynamic programming, Divide and Conquer [Merge Sort] ,Greedy [Huffman code has been asked many times for GATE] and Brute Force.
- Practice basic problems like quick sort, merge sort, knapsack problem, matrix chain multiplication, LCS, Job sequencing, Compressing Mechanism.
- Questions come from filling of hash tables with : Linear probing, Quadratic probing,
- Expected no. of empty slots after x insertions (application of probability), Load factor.
- Closed hashing, Property of a hash function and Universal Hashing

RESOURCES :

C PROGRAMMING ONE SHOT :

https://www.youtube.com/live/sVpUHcBWzvQ?si=fNKGgqQjldQ_0s4w

COMPLETE FULL PLAYLIST :

https://www.youtube.com/playlist?list=PLOG_8OIGMp73IsUhx_1i6nnS0RPgetrbG

FOR ALGORITHMS , go follow Striver , Neetcode , do Leetcode

If you have less time , and wanna just study it for the sake of GATE :

https://youtube.com/playlist?list=PLOG_8OIGMp71sEpjL2T7p8eAvuScptLWI&si=t4ZlqAqioEvFr3Q5

DATA STRUCTURES 👍

ONE SHOT : <https://www.youtube.com/live/LYek8Rhl-o0?si=MNzCZd2ffh-2F4Ug>

TREES -

1. https://www.youtube.com/live/jFbpavkuF6A?si=Tuu7kS4_Zt9WMLFL
2. <https://www.youtube.com/live/bj3Z5SPF70I?si=yfFackkaEA6yzv9d>
3. <https://www.youtube.com/live/LYek8Rhl-o0?si=MNzCZd2ffh-2F4Ug>

RESOURCES 👍

1. **DISCRETE MATHEMATICS AND ENGINEERING MATHEMATICS [COMPREHENSIVE SUBJECT] :**

A. Go through the **ONE SHOT lectures** first :

DISCRETE : <https://youtu.be/h4sd7wRcyR0?si=YzMR77Sx46vEXWm5>

ENGINEERING MATH:

LINEAR ALGEBRA -

- i. <https://youtu.be/-QnxKVzQJ-I?si=1HJXgl4QscbEEePh>
- ii. <https://youtu.be/qnaLQyLQbFg?si=3GUSbO638SRCy9aF>

B. **understand the topic more from here** [THIS IS A COMPREHENSIVE SUBJECT] :

i. ENGINEERING MATH

https://youtube.com/playlist?list=PLvTTv60o7qj_tdY9zH7YceES7jfXiZkAz&si=7L0-C5m2yE-4QAwL

ii. DISCRETE MATHEMATICS :

<https://youtube.com/playlist?list=PLC36xJgs4dxEYmhZVBW7nBdftFZ4xmiF1&si=p6srORatTQgWCINU>

B. Start **practising questions** . Practise PYQ's , each PYQ for 4 - 5 times atleast

i. Keneth Rosen book (Pdf available online for free)

ii. **What to put more importance in :**

Types of Problems • Simple problems on logic. Sets Related Questions. Properties of relation, function. Partial and Total Ordering. • Hasse Diagrams, Group Theory. • Different types of graph and their properties : vertex and edge connectivity , separable graph , k-connected graph, connected component, matching , graph coloring (4-color theorem), Euler and Hamiltonian Graphs, Konigsberg Bridge problem, Independent set of vertices, chromatic number are important. • Tricky Problems on Permutations and Combinations. Pigeonhole principle. • LA : Eigen values and eigen vectors question. Simple questions related to matrix. Finding Values of variable with some properties of linear equations like infinite no of solutions or unique solution. • Calculus : Finding Maxima and Minima. Properties of limit, Continuity and differentiability. Finding values by Mean Value Theorem. Integration. • Probability : practice Bayes theorem, Normal and Poisson distribution, mean and variance of different distributions, standard deviation.

Topics and Chapters :

- Propositional and first order logic - Rosen Chapter 1 - 1.1 to 1.6
- Propositional Logic - Page 1 to 12. page 16 and 17
- Propositional Equivalences - Page 25 to 31
- Predicates and Quantifiers - page 37 to 49
- Nested Quantifiers - page 57 to 63
- Rules of Inference - page 69 to 78
- Sets - Rosen Chapter 2 : 2.1 , 2.2 Relations - Rosen Chapter 9 , 9.1 to 9.5
- Functions - Rosen Chapter 2.3
- Partial orders and lattices - Rosen Chapter 9.6
- Groups : Group, Abelian Group, Semigroup, Monoid, Ring, Integral domain fields from IITM lectures 35, 36, 37
- Graphs : connectivity, matching, coloring is there in syllabus
 - Rosen - connectivity, euler and hamilton paths , coloring is in chapter 10 – 10.4, 10.5, 10.8
 - Narsingh Deo Chapter 2 - 2-5, 2-6, Chapter 4 - 4-5 [connectivity], Chapter 8 - 8-1, 8-2, 8-4, 8-6.
- Counting, recurrence relations, generating functions - Rosen Chapter 6, 8
- Linear Algebra : Kreyszig - 9th edition Chapter 7.1 to 7.7, 8.1 to 8.3, video lectures

- Calculus : mean value theorem page 402 Kreyszig , and also from above video link
- Probability : Sheldon Ross,8th edition, chapter 1, 2, 3, 4 [exclude 4.8] ,5 [exclude 5.6]

Types of Problems

- Simple problems on logic. Sets Related Questions. Properties of relation, function. Partial and Total Ordering.
- Hasse Diagrams, Group Theory.
- Different types of graph and their properties : vertex and edge connectivity , separable graph , k-connected graph, connected component, matching , graph coloring (4-color theorem), Euler and Hamiltonian Graphs, Konigsberg Bridge problem, Independent set of vertices, chromatic number are important.
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- Calculus : Finding Maxima and Minima. Properties of limit, Continuity and differentiability. Finding values by Mean Value Theorem. Integration.
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ENGINEERING MATHEMATICS PYQ SERIES and REVISION lecture :

https://youtube.com/playlist?list=PLs5_Rtf2P2r4Nya_2IG_RZ9nknAAQj4rz&si=smEZEYYv_uboOJvr

II.THEORY OF COMPUTATION

Types of Problems:

Finite Automata covers approximately 50% questions from TOC. This topic is very important

- Minimization of finite Automata , Closure Properties of finite automata .
- Finding minimum number of states, NFA to DFA conversion, Finding Regular Expressions, Mealy Moore machine .
- Regular Expression identification.
- Construction of finite Automata from regular Expression
- Generation of regular expression from finite Automata
- Equivalence of regular Expression

CFG and PDA : Practice more on problems on simplification of CFG, pushdown automata,

closure properties etc.

Regular and Context Free Languages :

- Closure properties of regular language

- Decidability of regular language
- Check whether the given language is regular or not
- Do problems on finding category of any language or grammar.
- Concepts related to expressive power of different languages.

Turing Machine and Undecidability :

- Basic problems related to NP-Completeness.
- Properties of Recursive and Recursively Enumerable Languages.
- Turing machine design and expressive power of different types of Turing machine.

RESOURCES :

TOC is practise heavy , so study accordingly

One shot lecture : <https://www.youtube.com/live/2MB6gFgS4vY?si=gNUWeBAgVW-U-95F>

Full comprehensive lecture : 2 videos / day speed to finish in 77 days

<https://youtube.com/playlist?list=PLC36xJgs4dxGvebewU4z2CZY0-8nB93E7&si=GiaTognDegZzRXq>

THIS IS A COMPREHENSIVE SUBJECT , SO YOU HAVE TO FINISH FULL LECTURE SERIES .

Time = July 15 to September 30 = 77 days

III.COMPUTER NETWORKS :

Types of Problems

- Addressing related questions : Subnet address, supernet address, broadcast address, range of network, no of host, classless addressing, non continuous addresses, first host and last host finding etc.
- Properties Of Circuit Switching and packet switching, Routing Protocols and Numerical Problems on them.
- Flow Control and Error Control Policies.
- Numerical Problems on Window Size [sliding window protocols] , No Of Sequence bits, frame size, bandwidth, round trip time, utilization, Hamming Distance, CRC.
- Congestion Control policies like slow start, congestion avoidance and Congestion Detection.
- IP Header , TCP and UDP header format, theory related to Ethernet and token ring.
- Basics Of Diferent Type of protocols like : FTP, HTTP, DHCP, ARP, RARP, SMTP, ICMP, POP.

- Basic Concepts of Cryptography and firewalls.

RESOURCES :

One shot lecture : <https://www.youtube.com/live/4xjwwwF8P7Y?si=Nl-x7eELiYVPBOWf>

Full series : 202 videos : 2 lectures / day

https://youtube.com/playlist?list=PLC36xJgs4dxHT-TxTy3U1slr5RaBJGaLd&si=1I7VSnG5kt_EtPq2

Time = 108 days

THIS IS A COMPREHENSIVE SUBJECT , SO YOU HAVE TO FINISH FULL LECTURE SERIES .

Time = July 15 to October 31

IV. COMPUTER ORGANISATION AND ARCHITECTURE :

Types of Problems

- Addressing Modes : Theory and questions
- Numerical related to program counter after some instruction, no of one address-two address instructions, Values after shift and rotate instructions, horizontal and vertical programming related questions.
- Numerical Problems on Speed up of pipeline, time taken to complete instruction in pipeline and non-pipeline architectures, Hazards in pipeline, hazards removal, branch penalty etc.
- Numerical Problems on cache memory organization, mapping technique, multilevel caches, write through and write back technique

RESOURCES :

One shot lecture : 10 videos series

<https://youtube.com/playlist?list=PL3eEXnCBViH8OS7fH0uQdre5YGCIOhCBH&si=v92wvk6VOntvPXIW>

Full series : 200 lectures : 3 lectures / day

https://youtube.com/playlist?list=PLBlnK6fEyqRgLLlzdgiTUKULKJPYc0A4q&si=K0tVgQamCXa_ICvp

Time = 75 days

THIS IS A COMPREHENSIVE SUBJECT , SO YOU HAVE TO FINISH FULL LECTURE SERIES .

Time = July 15 to September 30

V. OPERATING SYSTEM :

Types of Problems

- **Scheduling** : Numerical Questions have more chances. Practice more in finding turn around time and waiting time of different scheduling policies.
- **Deadlock** : Bankers Algo, Given Sequence is safe or not. Chances of common data or linked questions.
- **Concurrency and Synchronization** : High Probability Of Questions in exam.
- Practice some question related to semaphores and classical problems of synchronization (this will help you to solve other questions), Mutual Exclusion case using P and V , Critical section problem.
- **Memory Management** : Questions generally comes from page table size, number of pages, logical address, physical address, page size, inverted page table, virtual memory, TLB etc.
- **File systems**: Algorithms for disk scheduling

RESOURCES :

One shot lecture : https://www.youtube.com/live/86kN-pdxsl0?si=x1LhXcRXb_v70m34

Small lecture playlist : 7 videos series

<https://www.youtube.com/live/1RQrCXiPME8?si=xShb8XaAX-ahAfVC>

Full series : 62 lectures : 1 lecture / day

<https://youtube.com/playlist?list=PLG9aCp4uE-s17rFjWM8KchGlffXgOzzVP&si=A5HNNViGiiXRfSvu>

Time = 61 days

THIS IS A COMPREHENSIVE SUBJECT , SO YOU HAVE TO FINISH FULL LECTURE SERIES .

Time = September 1 to October 31

VI. DATABASE MANAGEMENT SYSTEMS :

Types of problems

ER-model and Relational model : relational algebra, tuple calculus

- Minimum number of tables required [2005, 2008]
- Properties of ER model [2012]
- Meaning of given Tuple Relational Calculus query [2008, 2007]
- Normal form [2016]
- Meaning of relational algebra statement [2007, 2014]
- Functional dependencies [2005]
- Relation algebra optimized version [2014] , problems on Join [2004, 2014] , optimized form [2014], no.of tuples [2012 , 2013]
- Meaning of query [2008]

SQL : Practice select clause properly with additional properties of having, group by, any, all, exit. Question may come with relational algebra in common data section.

Integrity constraints, normal forms.

- Normalization : find normal form [2016 - 09, 08, 03], finding candidate keys [2005, 2011, 2013, 2014] , decomposition of relation [2002] , loss less join and dependency preservation [2001] minimal cover [2014]

Transactions and concurrency control

- Finding View and Conflict serializability [2003, 2008, 2010, 2014, 2012]
- Finding Recoverable [2014, 2006 ,2015 ,2016] and Cascade schedule.
- Lock based[2004], Two phase, time stamp and graph based protocol with their properties like deadlock freedom [2016], starvation freedom.
- Acid properties in real life situation in transaction [2015 , 2016]

File organization and indexing.

- Formation and structure of B and B+ trees
- Primary and clustering index[2008 , 2002, 2013, 2015]
- Numerical Questions from no of block required in indexing of different type,[IT 2005]
- collision resolution minimum and maximum no of nodes in B,B+ trees [2010].
- ordered indexing & hashing 2011

RESOURCES :

One shot lecture :

PART 1 : https://www.youtube.com/live/Om4W_hBr1hg?si=jAiD2e9DoH_pzn3l

PART 2 : <https://www.youtube.com/live/dhWcyGB5spQ?si=sQbTom9vugcwz127>

Small lecture playlist : 10 videos

https://youtube.com/playlist?list=PLOG_8OIGMp719MUzCtMjbngggqwmZYdOXg&si=eDzR3yDH9fYuKtXT

Full series : 225 lectures : 3 lecture / day

<https://youtube.com/playlist?list=PLC36xJgs4dxGcz7nZaxGxxmbJrcgDXhFk&si=f2B45FVFMbx0tzxj>

Time = 76 days

THIS IS A COMPREHENSIVE SUBJECT , SO YOU HAVE TO FINISH FULL LECTURE SERIES .

Time = October 1 to December 15

VII. COMPILER DESIGN

Previous year problems

- 3-address code : [minimum number of temporary variables] constructing 3 address code for an expression (See Topic 6.2)
- Abstract Syntax tree - from syntax directed translation (Topic 5.3)
- Control flow graph no of nodes and edges from Intermediate code generation (Topic 6.6) [CFG not in syllabus]
- Finding First and Follow (Topic 4.4)
- Parsing : There is always a question related to parsing. You need to practice all parsing technique because there is also chances for linked questions. (Topics 4.4-4.9)
- Finding internal node in syntax-directed translation 5.4, 5.5
- Number of token generated 3.3 ,3.4
- Questions from Topics 4.4 – 4.7 LL(1)-LR(1)-SLR-LALR-CLR.
- Precedence and Associativity of operators. (Topics 4.8, 4.9)
- Finding value from expression tree. (Topics 2.8 , 6.1)
- Ambiguous grammar (Topic 4.3)

Just this playlist . To be completed in 15 days only .

https://youtube.com/playlist?list=PLXFzGMAWxbdcBXrdYwYiDu3sIFYhUD_cX&si=ljvqFFFGNQpqhrfY

If you still don't have time :

<https://www.youtube.com/live/nt3MlkpmUCw?si=brKE0x7ksXyq5XkL>

Its a single 8 hour long lecture

VII. DIGITAL LOGIC :

ONLY THIS

<https://youtu.be/IH0sYax5Yg0?si=dfhiBGld2xI2fwFy>

TOPIC :

Topics :

Boolean algebra : Laws of Boolean algebra, Theorems of Boolean algebra, Switching functions, Methods for specification of switching functions - Truth tables and Algebraic forms, Realization of functions using logic gates.

Combinational and sequential circuits :

I. Design of Combinational Logic Circuits:

- Gate level design of Small Scale Integration (SSI) circuits, Modular combinational logic elements - Decoders, Encoders, Priority encoders, Multiplexers and Demultiplexers.
- Design of Integer Arithmetic Circuits using Combinational Logic: (Application)
- Integer adders - Ripple carry adder and Carry look ahead adder, Integer subtractors using adders, Unsigned integer multipliers - Combinational array circuits, Signed integer multipliers - Booth's coding, Bit-pair recoding, Carry save addition and Wallace tree multiplier.
- Signed integer division circuits - Combinational array circuits, Complexity and propagation delay, analysis of circuits.

II. Sequential Circuit Elements:

Latches -RS latch and JK latch, Flip-flops-RS, JK, T and D flip flops, Master-slave flip-flops, Edge-triggered flip-flops.

III. Analysis and Design of Synchronous Sequential Circuits:

- Models of sequential circuits - Moore machine and Mealy machine
- Flip-flops - Characteristic table, Characteristic equation and Excitation table•
- Analysis of sequential circuits- Flipflop input expressions, Next state equations, Next state maps, State table and State transition diagram
- Modular sequential logic circuits - Shift registers, Registers, Counters and Random access memories

IV. Design of Arithmetic Circuits using Sequential Logic :

Serial adder for integers, Unsigned integer multiplier, Unsigned integer division circuits, Signed integer division, Floating-point adder/subtractor - Design of control circuit, Floating
- point multiplier

V. Introduction to digital computer :

- Design of Arithmetic circuits – Adders, Multipliers
- Design of Memory – ROM/RAM
- Minimization (Simplification of Boolean Expressions and Functions) :
- Algebraic methods, Canonical forms of Boolean functions, Minimization of functions

using

- Karnaugh maps, Minimization of functions using Quine-McClusky method.
- Number representations and computer arithmetic (fixed and floating point) :
- Number systems and codes - (Binary, octal and hexadecimal number systems; Methods of base conversions; Binary, octal and hexadecimal arithmetic)
- Representation of unsigned and signed integers, Fixed-point representation of real numbers, Floating-point representation of real numbers

Types of Problems

- Practice K-Map minimization. SOP, POS forms and Don't care representation
- Practice questions related to Floating Point representation, integer representation,
- IEEE format, range and precision.
- Determining minimum number of NAND/NOR gates required to realize a boolean expression [2004, 2009]
- For a given truth table, find the function
- Important flip flops - J-K, D, T, R-S type of flip flops. For given flip-flops, modulus of counter is asked.
- Questions on J-K flip flop state sequence [2014, 2015]
- Calculate the propagation delay in flip flops
- 4 Input multiplexer 4 to 1, determine the output [2010,2014]
- For a given sequence, find out minimum number of j-k flip flop require to implement the counter [2016 , 2015] Questions on counters were asked in 2004, 2007, 2011, 2014
- Design of counter using flip flop [2015, 2016] Propagation delay of adder – [2004 - 62] , [2015 Set 1 – 47, Set 2 – 65]